

Lezione del 23-4-2026

giovedì 23 aprile 2026 09:09

la funzione $f(x) = ax^2 + bx + c$ $a \in \mathbb{R}$
 $a \neq 0$
 $b, c \in \mathbb{R}$

funzione QUADRATICA

Per me è noto il suo grafico
 una volta conosciuti i valori a, b, c
 Infatti me so il grafico di una

PARABOLA

che ha vertice

1

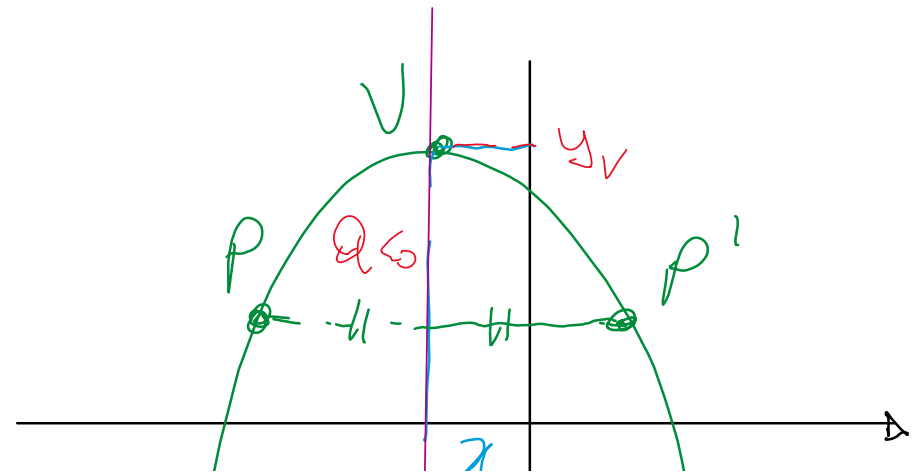
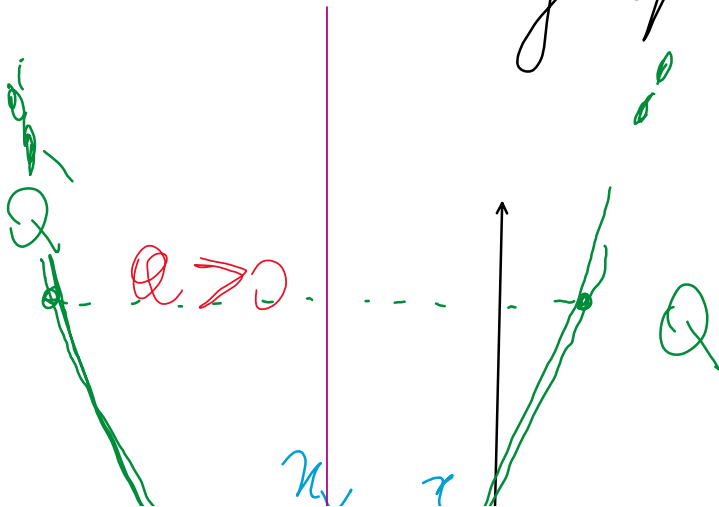
$$V = (x_v, y_v) \text{ dove } x_v = \frac{-b}{2a}$$

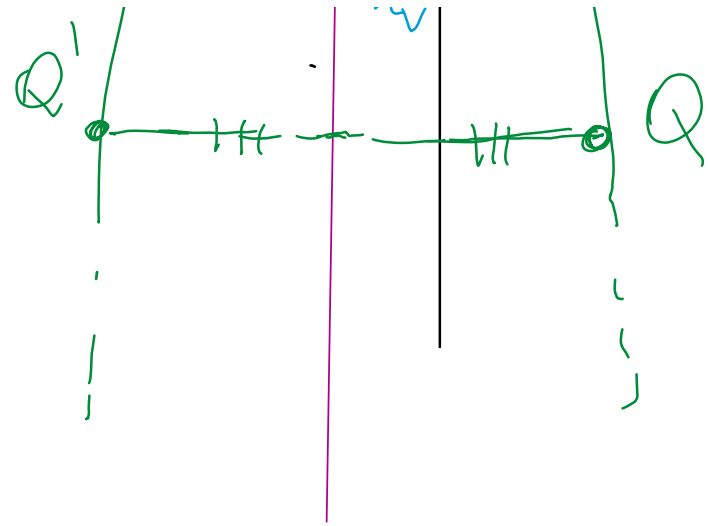
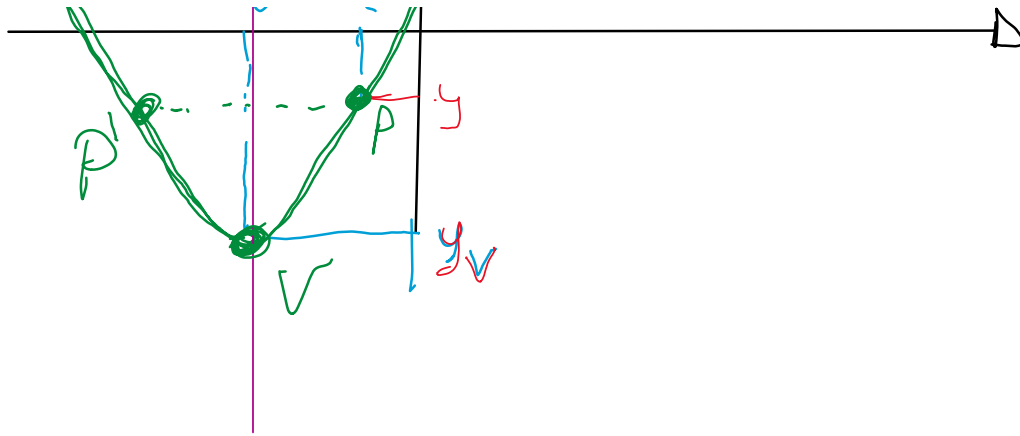
$$y_v = f(x_v) = -\frac{b^2 - 4ac}{4a} = -\frac{\Delta}{4a}$$

è ASSE DI SIMMETRIA la

retta (VERTICALE) $x = -\frac{b}{2a}$

Così un grafico del tipo





Ex

$f(x) = -2x^2 + 4x - 5 : \mathbb{R} \rightarrow \mathbb{R}$
 Chv il suo grafico:

$$V = (x_v, y_v) \quad x_v = -\frac{b}{2a} = -\frac{4}{2(-2)} = +1$$

$$y_v = f(1) = -2(1)^2 + 4(1) - 5 = -3$$

$$V = (1, -3)$$

$$x = +1$$

17 > 15
SINTETICA

Cerchiamo due punti del grafico (A Piacere)

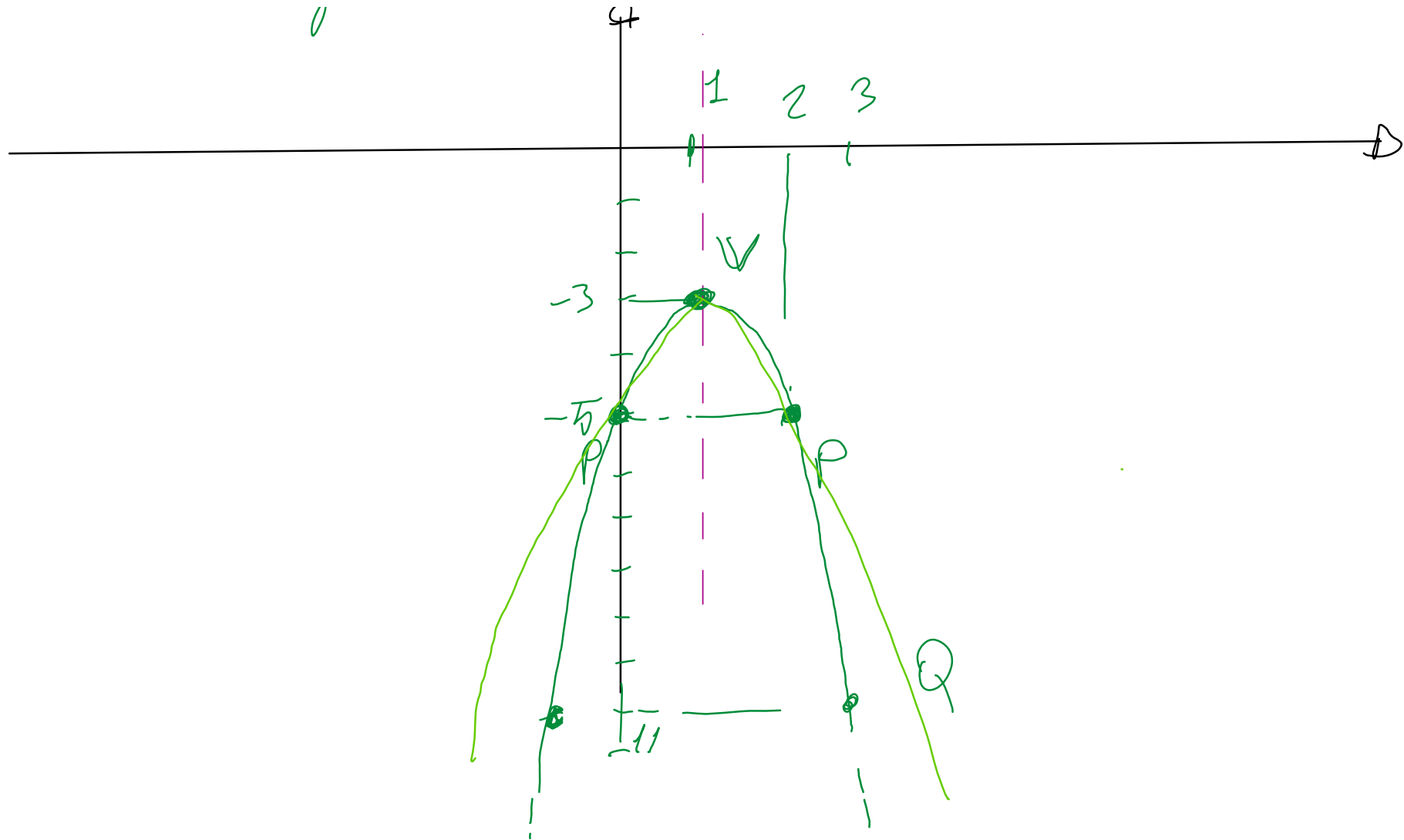
$$x = 0 \rightarrow f(0) = -2(0)^2 + h(0) - 5 = -5$$

$$P = (0, -5)$$

$$x = 2 \rightarrow f(2) = -2(2)^2 + h(2) - 5 = -8 + 8 - 5 = -5$$

$Q = (2, -5) = P'$
non è molto utile
ci fermi di disegnare
il grafico

$$x = 3 \rightarrow f(3) = -2(3)^2 + h(3) - 5 = -18 + 12 - 5 = -11$$



oss

Spesso una legge quadratica è
 molto utile

minima

$$f(x) = a(x - x_0)^2 + b$$

other $V = (x_0, b)$ è il vertice

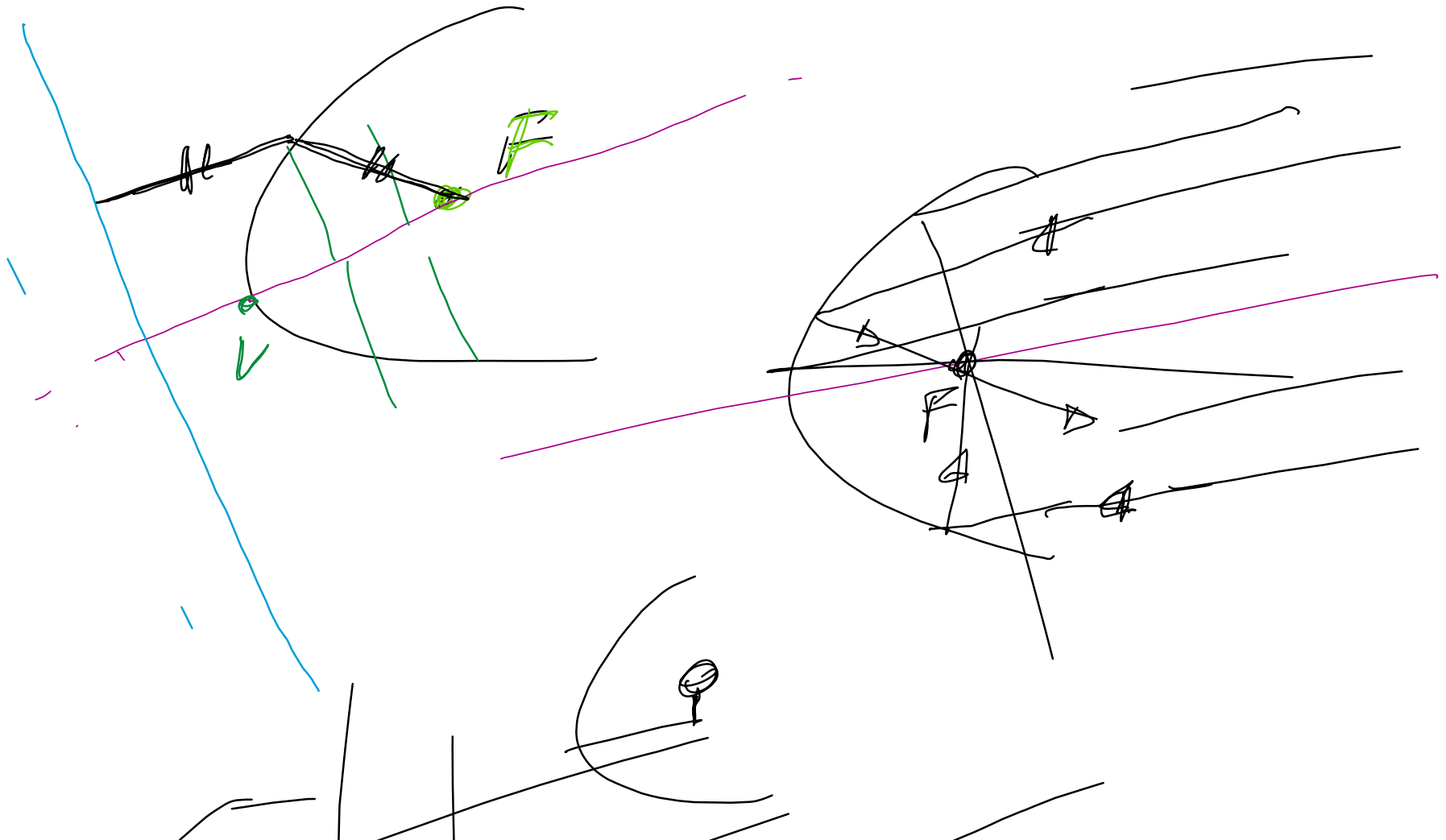
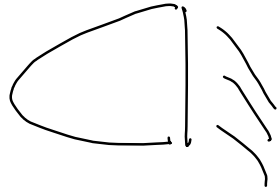
Ex $f(x) = 2(x - 3)^2 + 1$

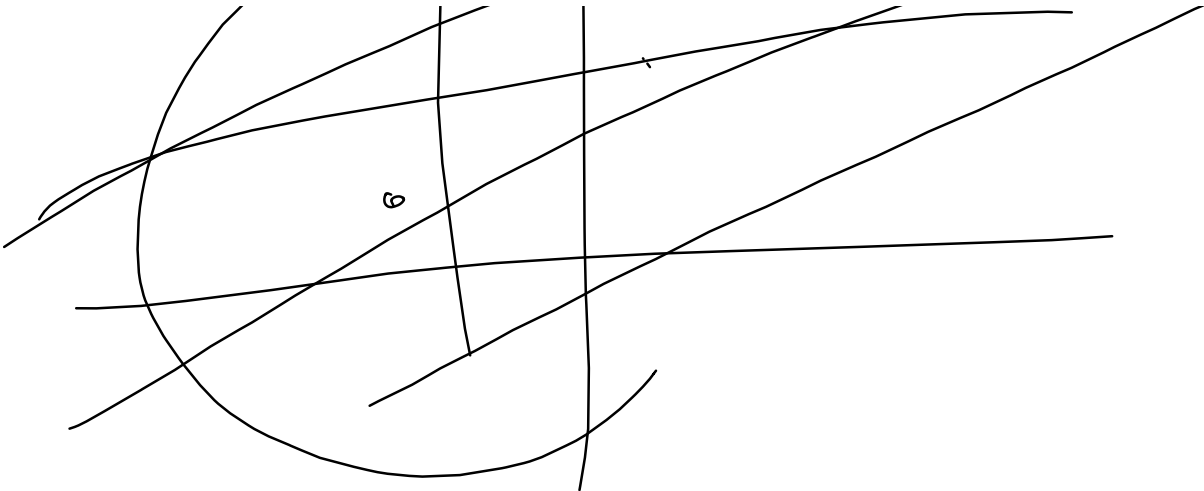
$V = (x_v, y_v)$ è dato subito da

$$x_v = -(-3) = 3$$

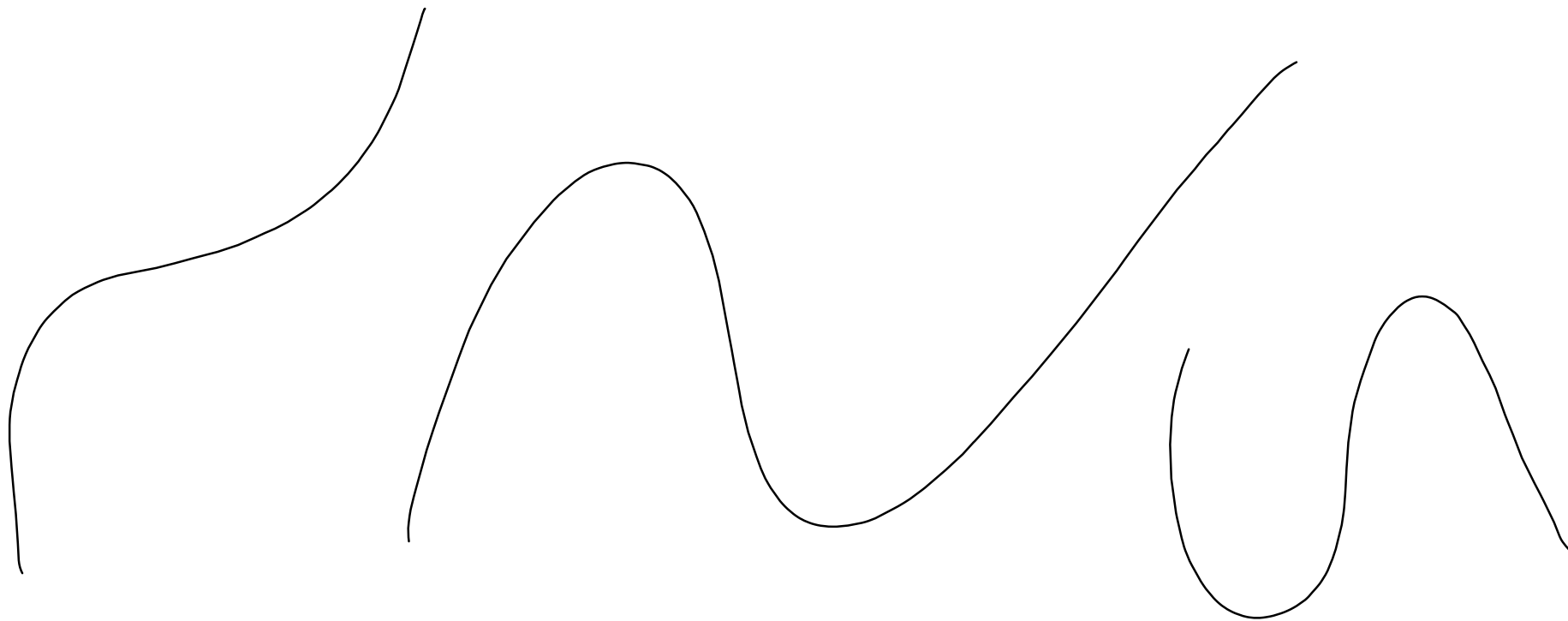
$$y_v = 1$$

$$V = (3, 1)$$





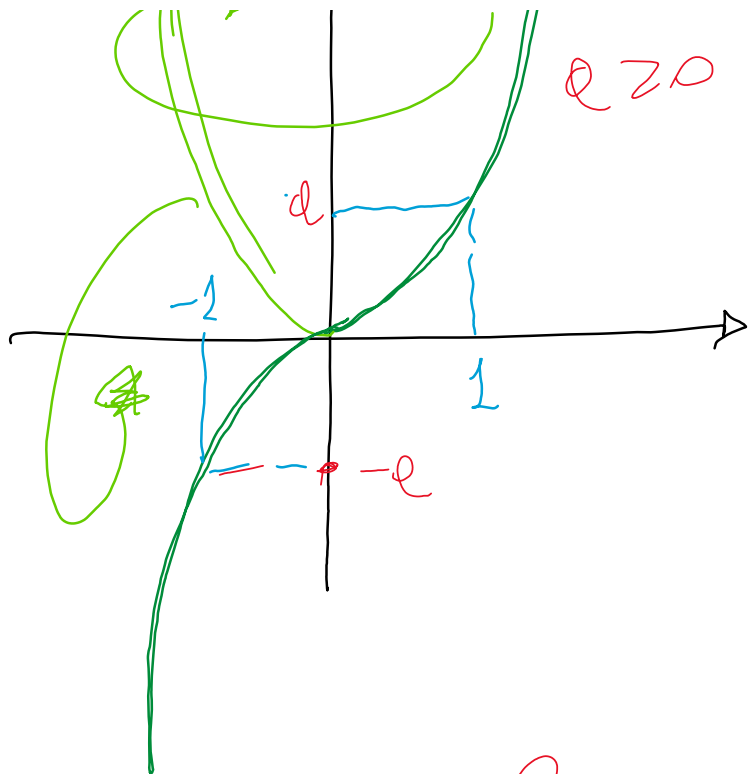
$$f(x) = ax^3 + bx^2 + cx + d : \mathbb{R} \rightarrow \mathbb{R}$$



$$f(x) = \textcircled{a} x^3 : \mathbb{R} \rightarrow \mathbb{R}$$

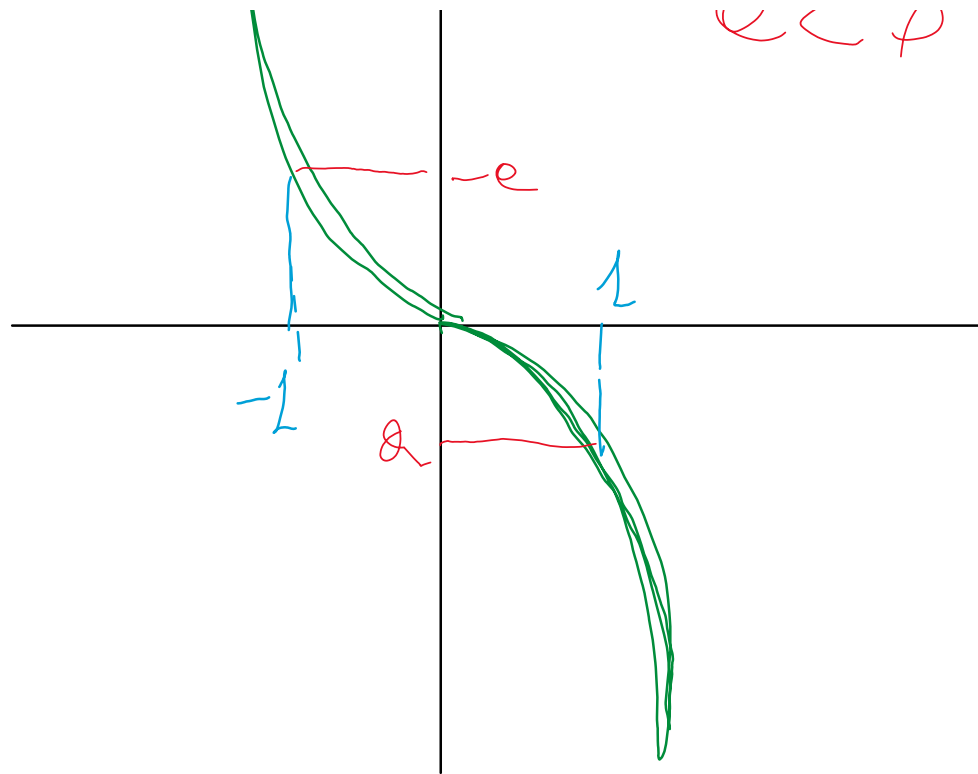
PARABOLA CUBICA





$$\text{Dom}(x^2) = \mathbb{R}$$

f è STROTTI. CRESC.
 f è INIETTIVA IN $\mathbb{R}$
 $Q \geq 0$



f è STROTTI. DECRESC.
 f è INIETTIVA IN $\mathbb{R}$
 $Q < 0$

più generale

$$f(x) = ax^n : \mathbb{R} \rightarrow \mathbb{R}$$

Se n è PARI il grafico di f
 è "SIMILE" al grafico delle parabole
 $g(x) = ax^2$

Se n è DISPARI è "simile" al grafico
 delle parabole cubiche $g(x) = ax^3$

Ex

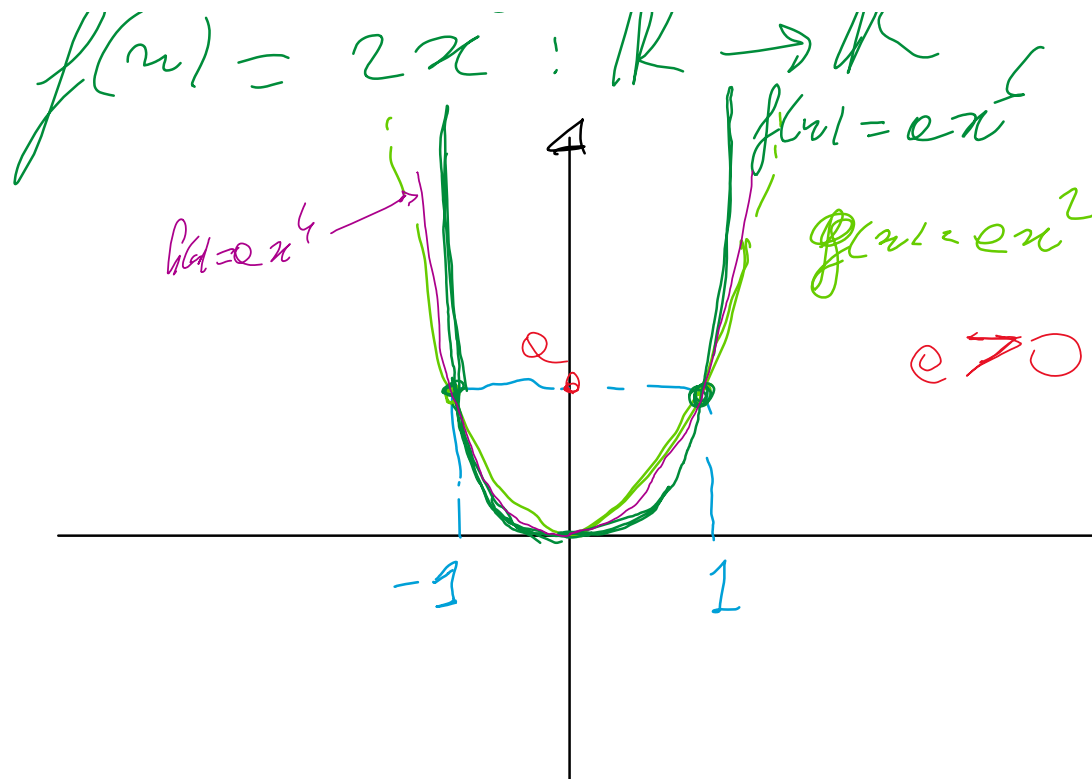
n

6

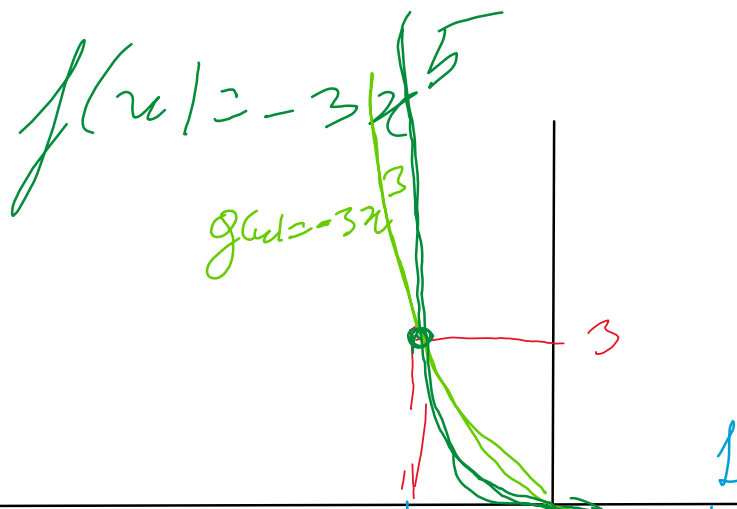
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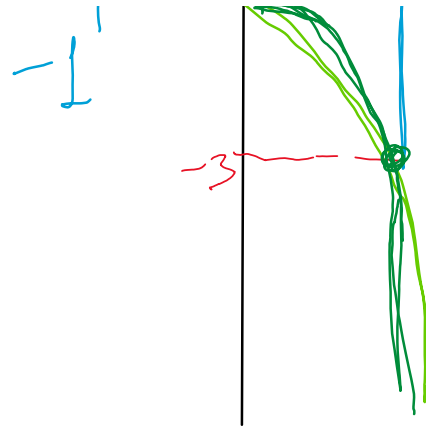
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Ex



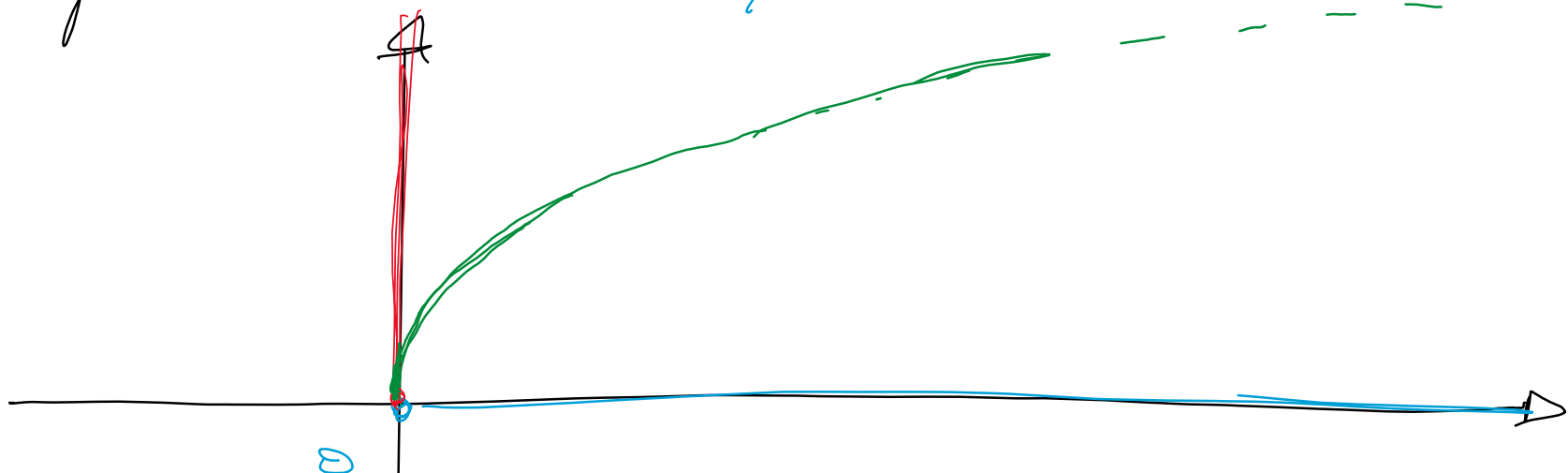
Ex





FUNZIONE RADICE QUADRATA

$$f(x) = \sqrt{x} : [0, \infty) \rightarrow \mathbb{R}$$



h. i) STR. COSTANTE

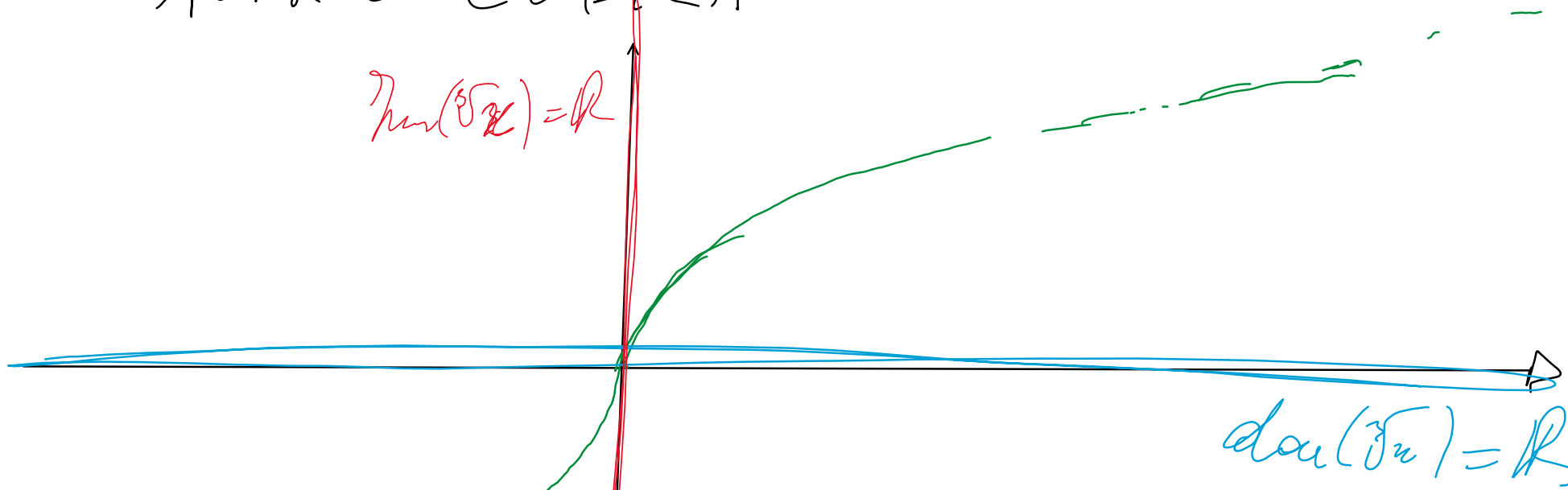
$$\text{Dom}(\sqrt{x}) = [0, +\infty[$$

(f $\bar{u}$ INIETTIVA)

$$f(x) = \sqrt[3]{x} : \mathbb{R} \rightarrow \mathbb{R}$$

RAZICE CUBICA

$$\text{Dom}(\sqrt[3]{x}) = \mathbb{R}$$



$$\text{Dom}(\sqrt[3]{x}) = \mathbb{R}$$

$$\dim(\mathcal{F}_n) = \mathbb{R}$$

$$\dim(\mathcal{F}_n) = \mathbb{R}$$

f è STRICCA. CRESCENTE IN $\mathbb{R}$
 f è INIETTIVA IN $\mathbb{R}$

$$f(n) = A^n : \mathbb{R} \rightarrow \mathbb{R}$$

esponenziale

~~4~~

~~1~~

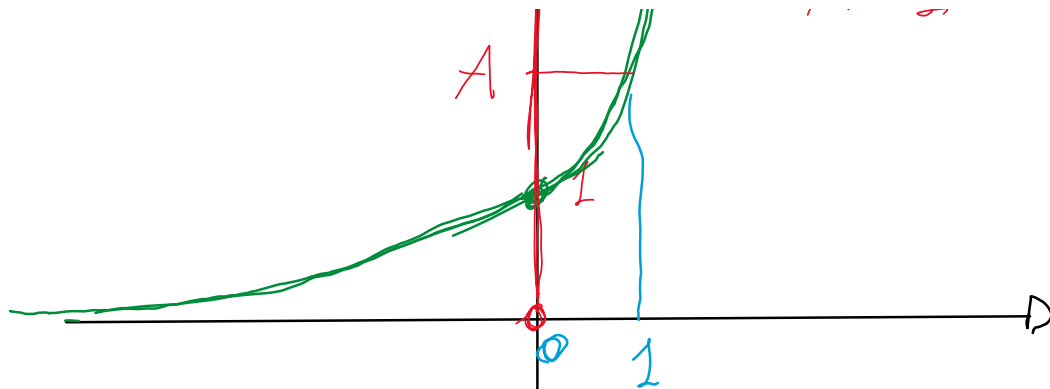
$A > 1$

$$A > 0$$

$$A \neq 1$$

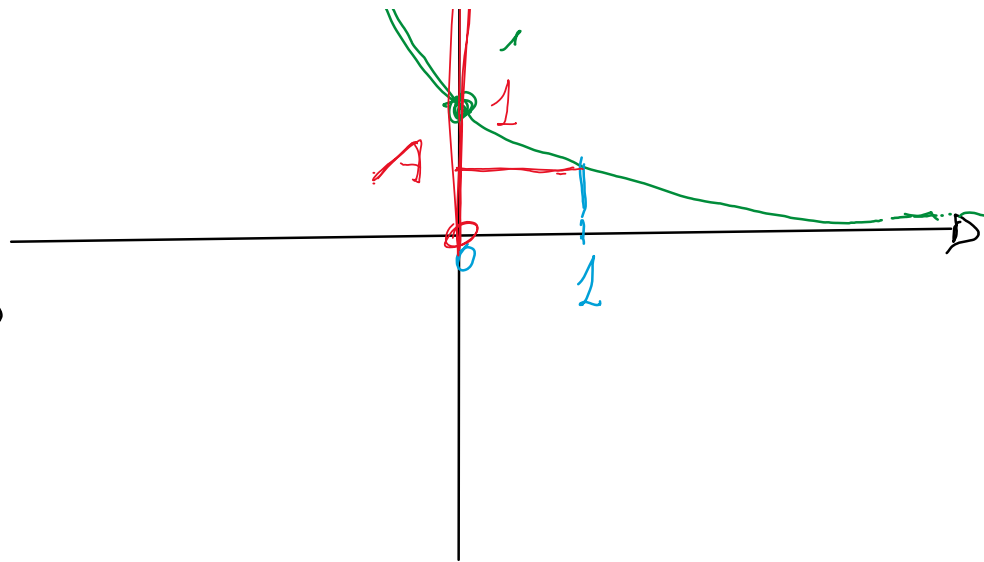
~~4~~

$$0 < A < 1$$



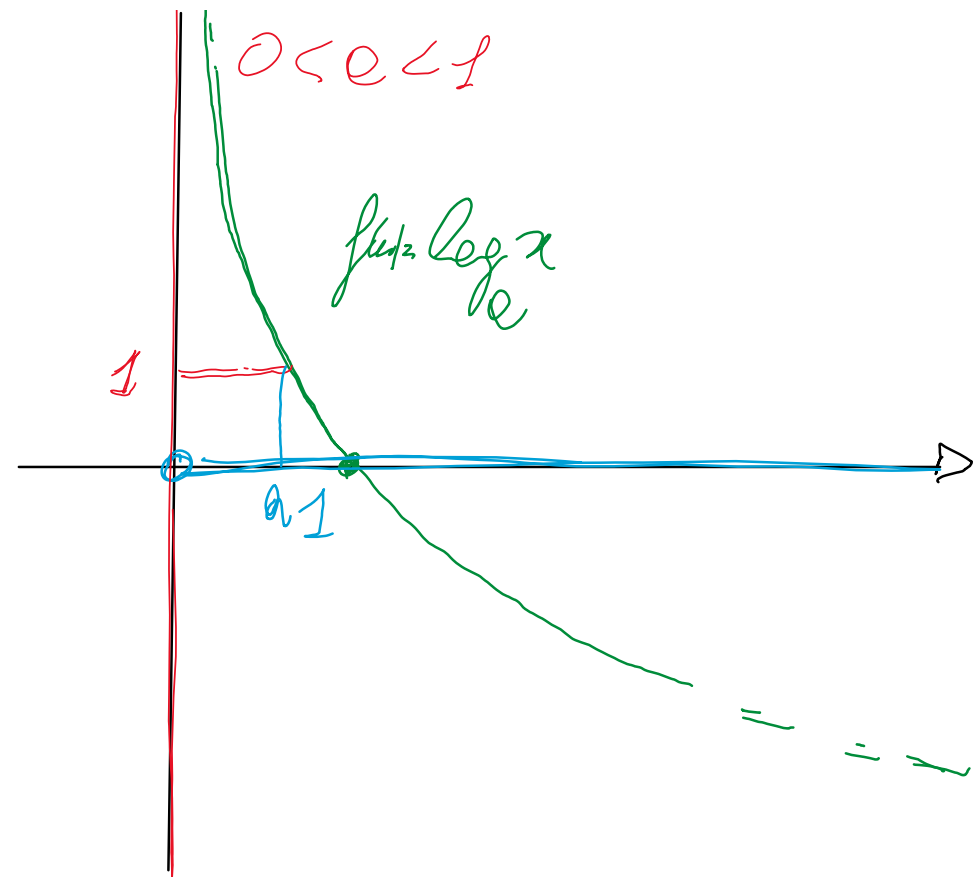
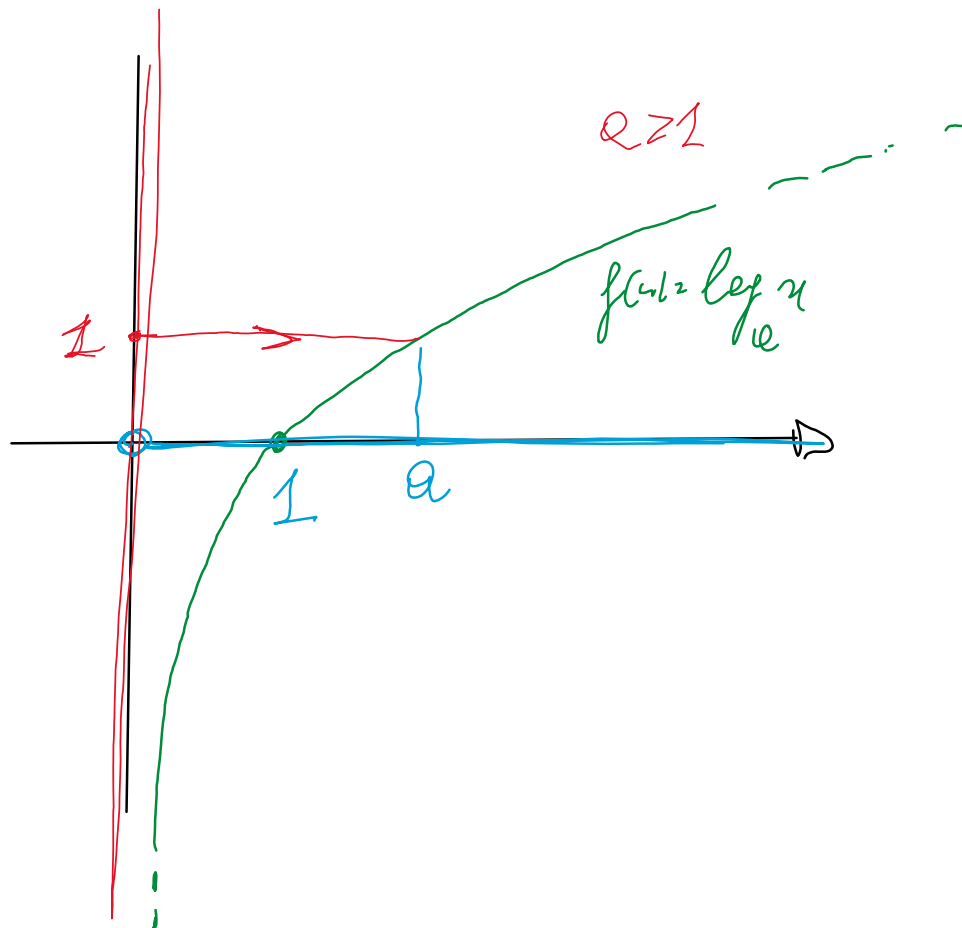
$$\text{Dom}(A^x) =]0, +\infty[$$

f strictly INCREASING in $\mathbb{R}$
(f is INJECTIVE in $\mathbb{R}$)



f strictly DECREASING

$$f(x) = \log_a(x) :]0, +\infty[\rightarrow \mathbb{R} \quad a > 0, a \neq 1$$



$$\text{Im}(\log_e x) = \mathbb{R}$$

$$\text{Dom}(\log_e x) =]0, +\infty[$$

f è STRETTA CRESCENTE IN $]0, +\infty[$

f è STRETTA DECRESCENTE

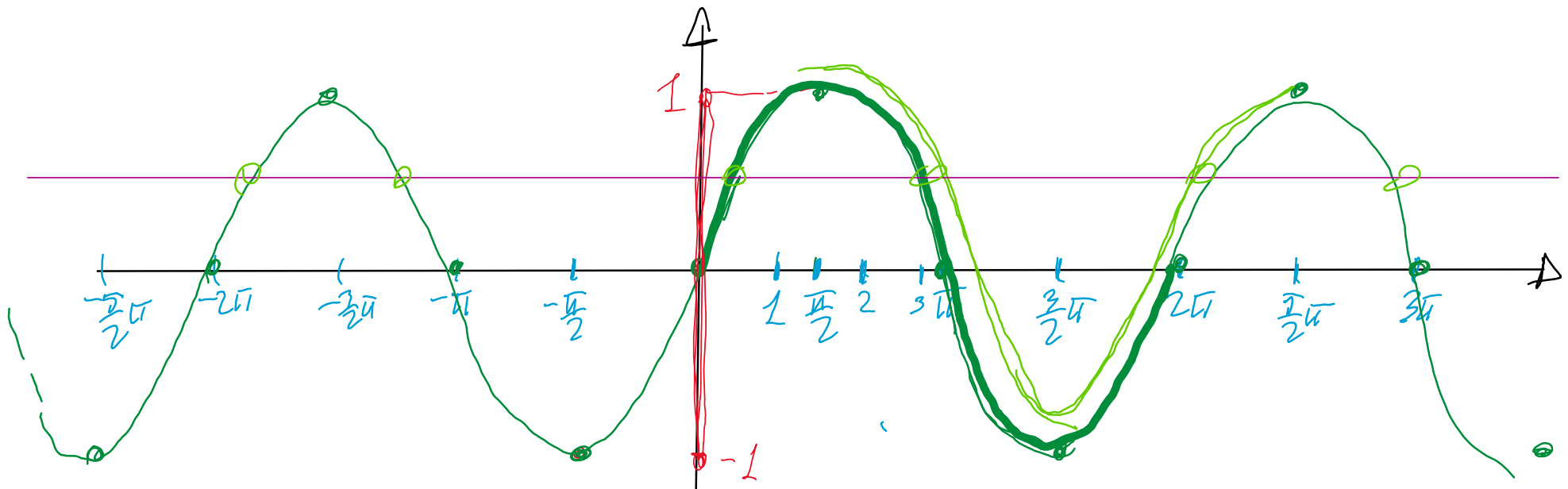
$$x \in \mathbb{R}$$

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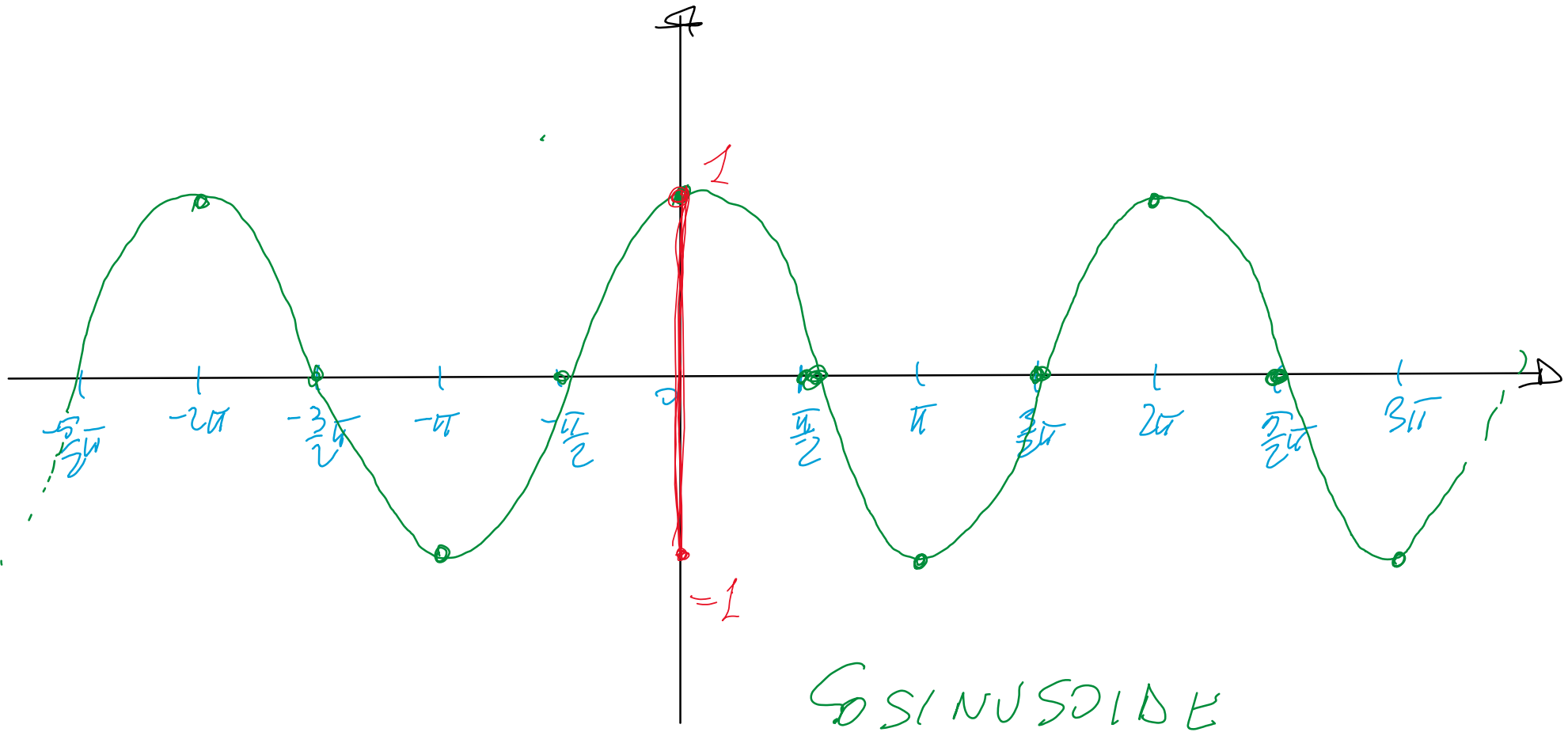
$$x \in \mathbb{R}$$

f. TRIGONOMETRICHE

$$f(x) = \sin(x) : \mathbb{R} \rightarrow \mathbb{R}$$



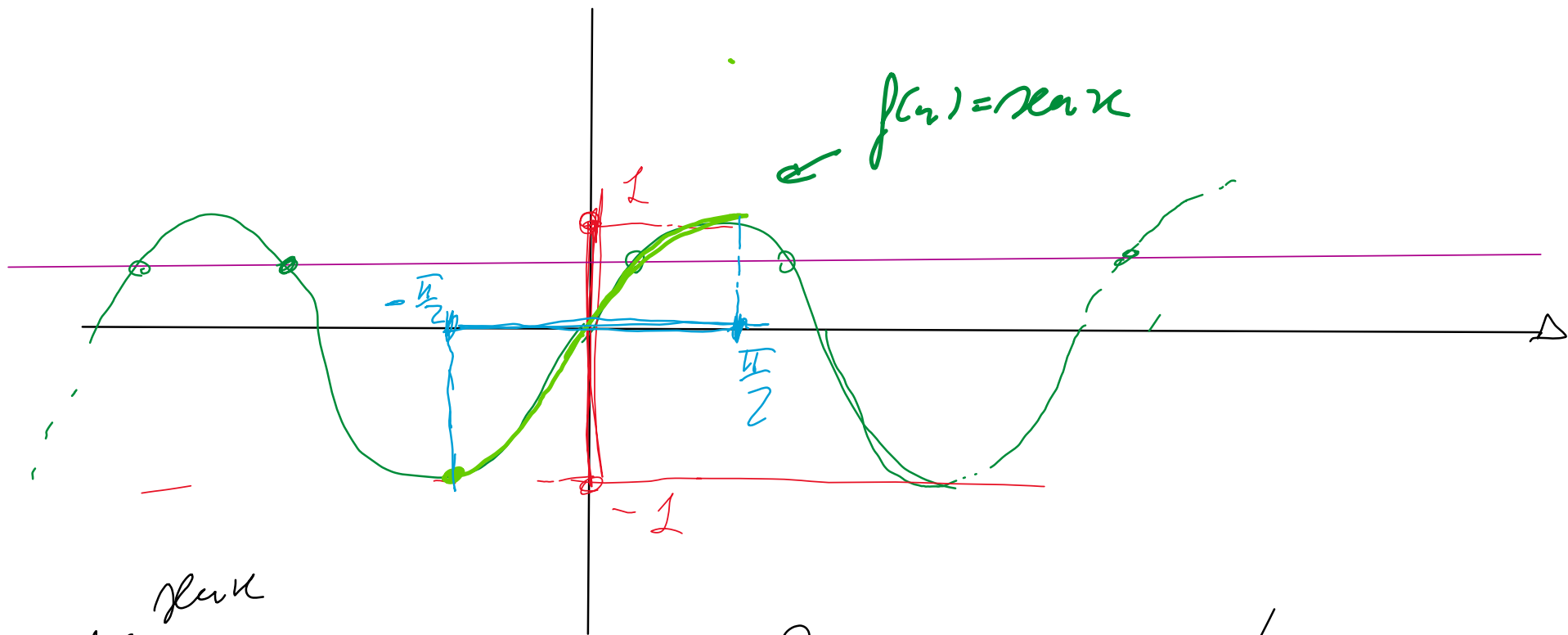
SINUSOIDE



$$\text{Dom}(\sin) = \text{Dom}(\cos) = [-1, 1]$$

f non $\in$ costante in $\mathbb{R}$

(Sei che se consideriamo INTERVALLI DI MONOTONIA STRETTA)



$f' = \cos x$
 f' non è monotona in $\mathbb{R}$, ma se restringiamo
 la funzione al solo dominio $X = [-\frac{\pi}{2}, \frac{\pi}{2}]$

Per trovare esiste la funzione inversa g

$$g: [-1, 1] \longrightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

con legge di definizione

$$[-1, 1] \ni y \longmapsto g(y) = x \text{ essendo}$$

x (l'unico elemento) di $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$\text{tale che } f(x) = \arcsin x = y$$

$$g\left(-\frac{\sqrt{3}}{2}\right) = x \text{ tale che}$$

$$\bullet x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\bullet \arcsin x = -\frac{\sqrt{3}}{2}$$

Angolo
di cui
l'angolo è

allora $x \begin{cases} \text{III} \\ \text{IV} \end{cases}$

$$\pi + \frac{\pi}{3} = \frac{4}{3}\pi + 2k\pi$$

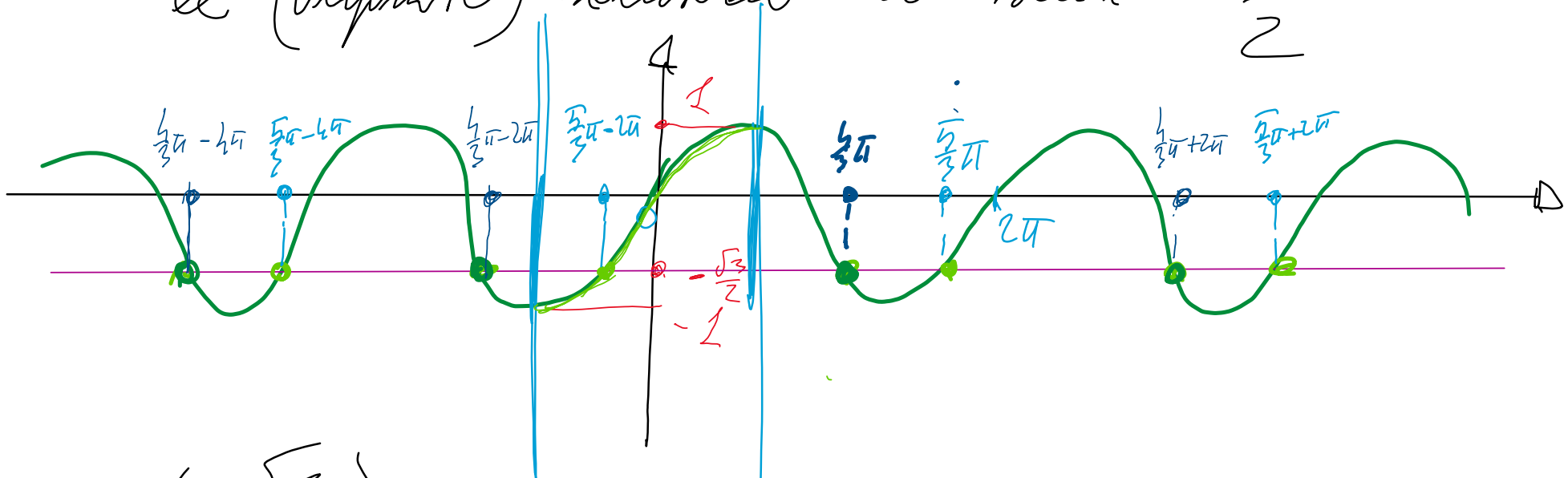
$$2\pi - \frac{\pi}{3} = \frac{5}{3}\pi + 2k\pi$$

Quadrante

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\alpha = \frac{\pi}{3}$$

le (infinite) soluzioni di $\sin x = -\frac{\sqrt{3}}{2}$



$$g\left(-\frac{\sqrt{3}}{2}\right) = \frac{4}{3}\pi - 2\pi = -\frac{2}{3}\pi$$

DEF

le funzione inverse di $\sin x: [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow \mathbb{R}$
 (ovvero di $\sin x$ RISTRETTA A $[-\frac{\pi}{2}, \frac{\pi}{2}]$)
 si dice ARC SENO ~~DEF~~

$g \equiv \arcsin(\cdot)$

$$\arcsin(y): [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$$

tale che

$$\arcsin(y) = x \quad \text{dove } x \in [-\frac{\pi}{2}, \frac{\pi}{2}] \text{ e}$$

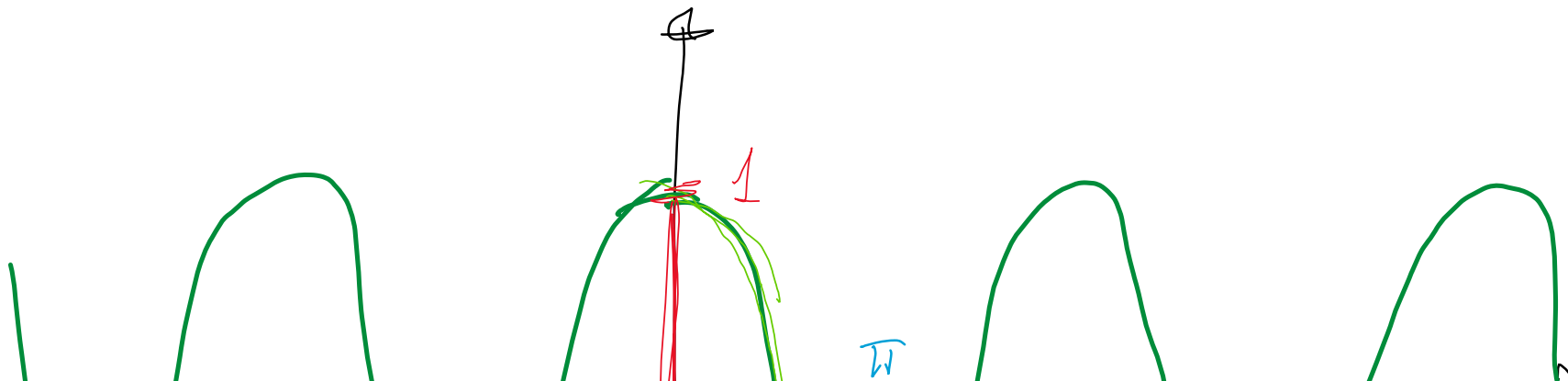
trovare $\arcsin x = y$

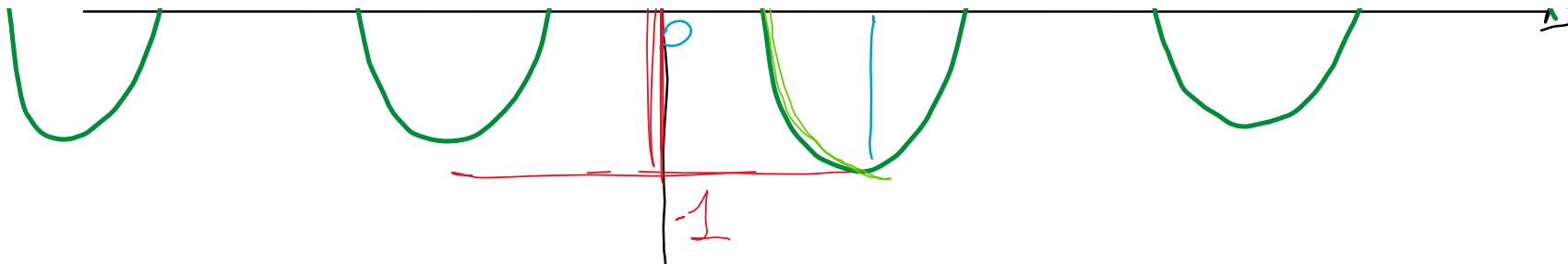
$$(\arcsin = \sin^{-1})$$

Calcolare

$$\arcsin(0); \arcsin(-\frac{1}{2}), \arcsin(2)$$

$$\arcsin(\frac{\sqrt{2}}{2})$$





arccoseno

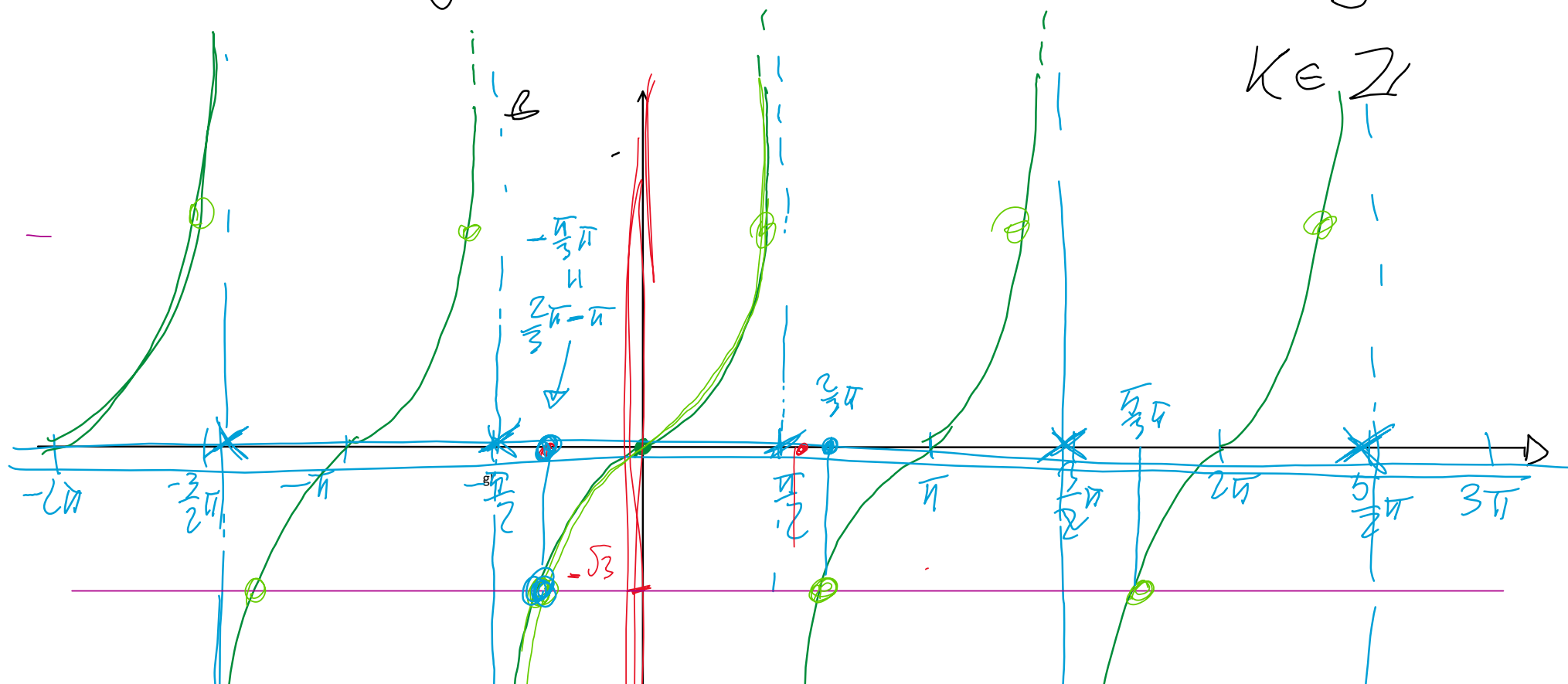
$$\arccos(y) : [-1, 1] \rightarrow [0, \pi]$$

$\arccos(y) = x$ dove x è (l'unico) valore
in $[0, \pi]$ tale che
 $\cos x = y$

$$\arccos\left(-\frac{\sqrt{3}}{2}\right), \arccos(-1)$$

$$\arccos\left(\frac{\sqrt{2}}{2}\right) - - -$$

$$\tan x = \frac{\sin x}{\cos x} : \mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi \right\} \quad k \in \mathbb{Z}$$



$$\sin(f(x)) \rightarrow \mathbb{R}$$

f non è MONOTONA in $\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\}$

f non è INIETTIVA

però ha intervalli
di MONOTONIA STRETTA

Considerate la restrizione di $f(x) :]-\frac{\pi}{2}, \frac{\pi}{2}[\rightarrow \mathbb{R}$
 $\downarrow$
 $] -\frac{\pi}{2}, \frac{\pi}{2}[$

dove $\tau_f x$ è STRUTT. (RISOLUZIONE)
 allora $\bar{x}$ potrebbe considerarsi la funzione
 inversa di $\tau_f x$
 $\downarrow$
 $]-\frac{\pi}{2}, \frac{\pi}{2}[$

$$f: \mathbb{R} \longrightarrow]-\frac{\pi}{2}, \frac{\pi}{2}[$$

$$f(y) = x \quad \text{dove } x \in]-\frac{\pi}{2}, \frac{\pi}{2}[\quad (\text{unico})$$

tale che $\tau_f x = y$

$$f(-\sqrt{3}) = x \quad : 1) x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$$

0

2) $\boxed{\tan \alpha = -\sqrt{3}}$

De informazioni sul quesito

è il valore
di tangente
in un angolo α
del I quadrante

$\tan \frac{\pi}{3} = \sqrt{3}!$

$\alpha = \frac{\pi}{3} = \pi - \frac{\pi}{3} = \frac{2}{3}\pi + k\pi$

$\frac{\pi}{3} = 2\pi - \frac{\pi}{3} = \frac{5}{3}\pi + k\pi$

$\frac{2}{3}\pi + k\pi$

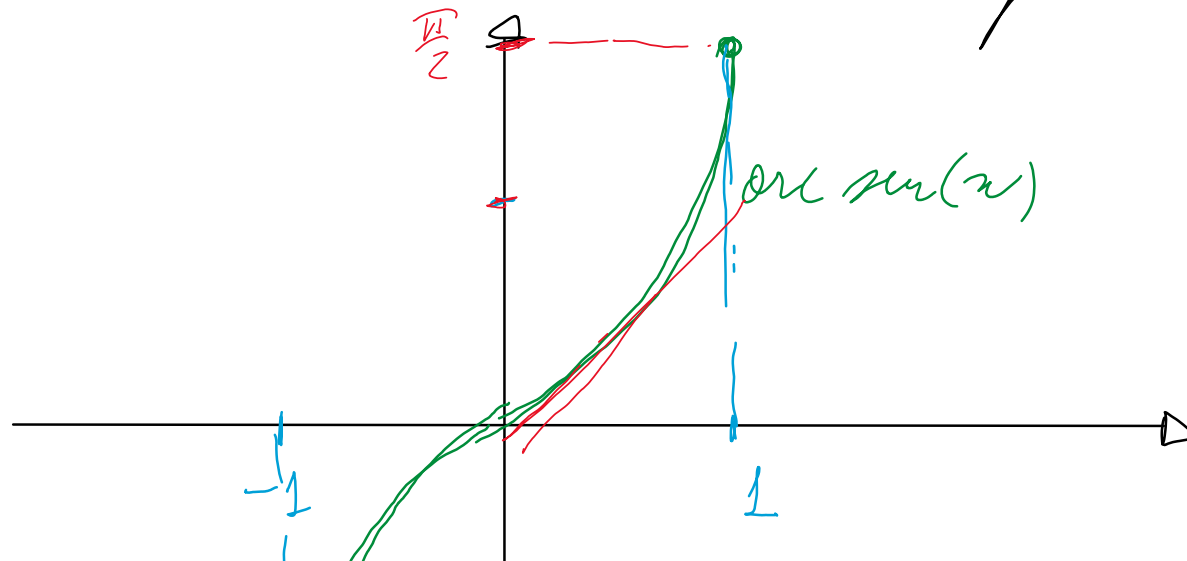
$0 - \frac{\pi}{3} = -\frac{\pi}{3} + k\pi$

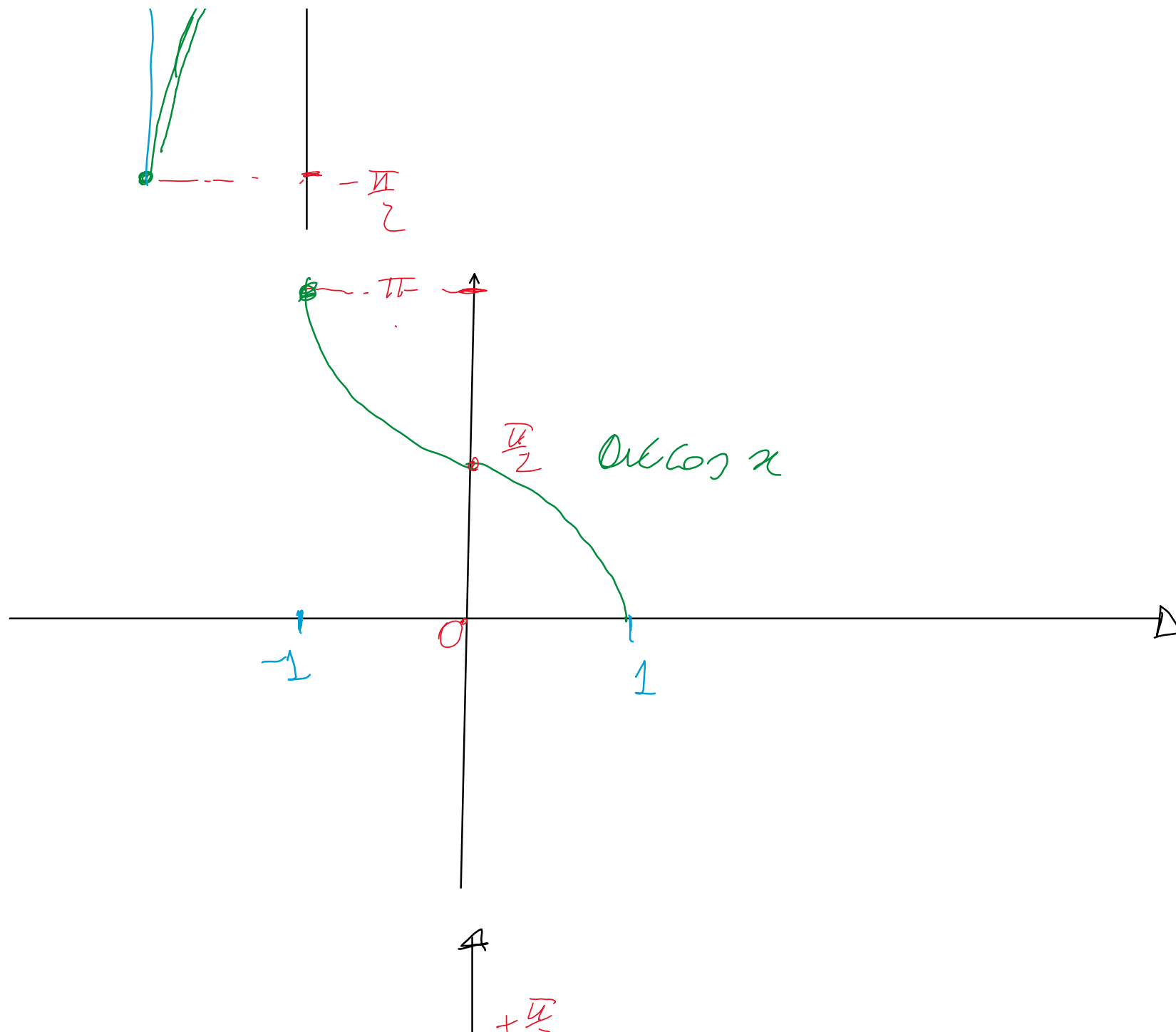
$g(-\sqrt{3}) = -\frac{\pi}{3}$

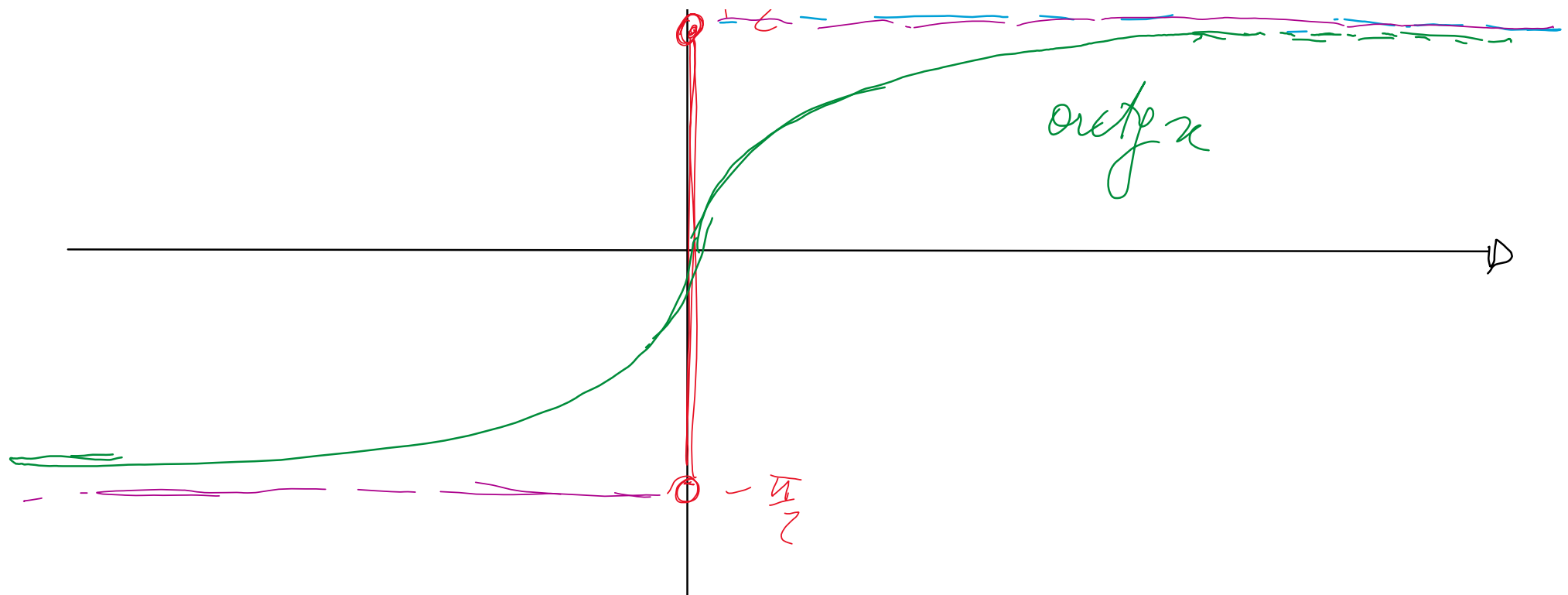
De informazioni inverse di $\tan \alpha$

π due ARCTANGENTE $\left] -\frac{\pi}{2}, \frac{\pi}{2} \right[$
 e π inoltre con
 arctg $\mathbb{R} \rightarrow \left] -\frac{\pi}{2}, \frac{\pi}{2} \right[$

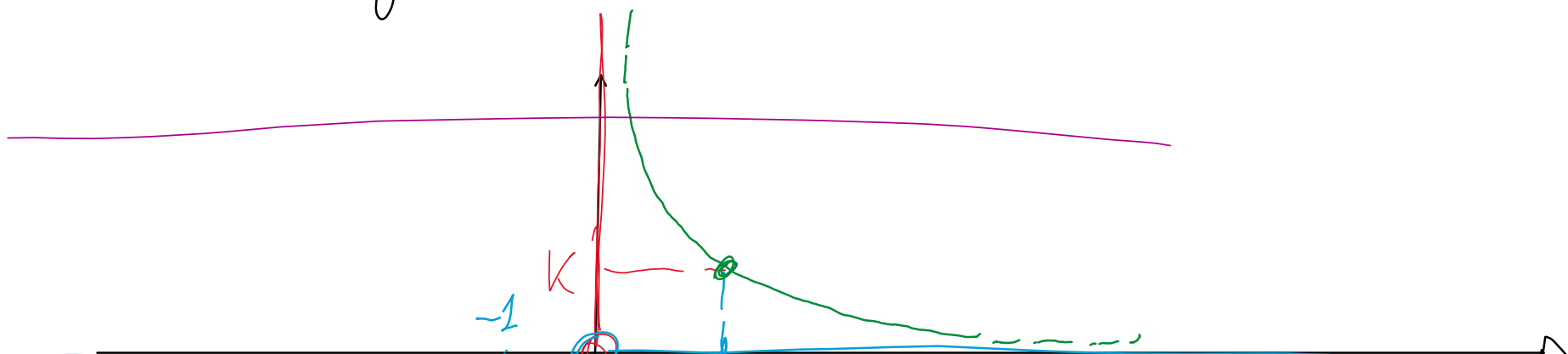
GRAFICI delle funzioni inverse

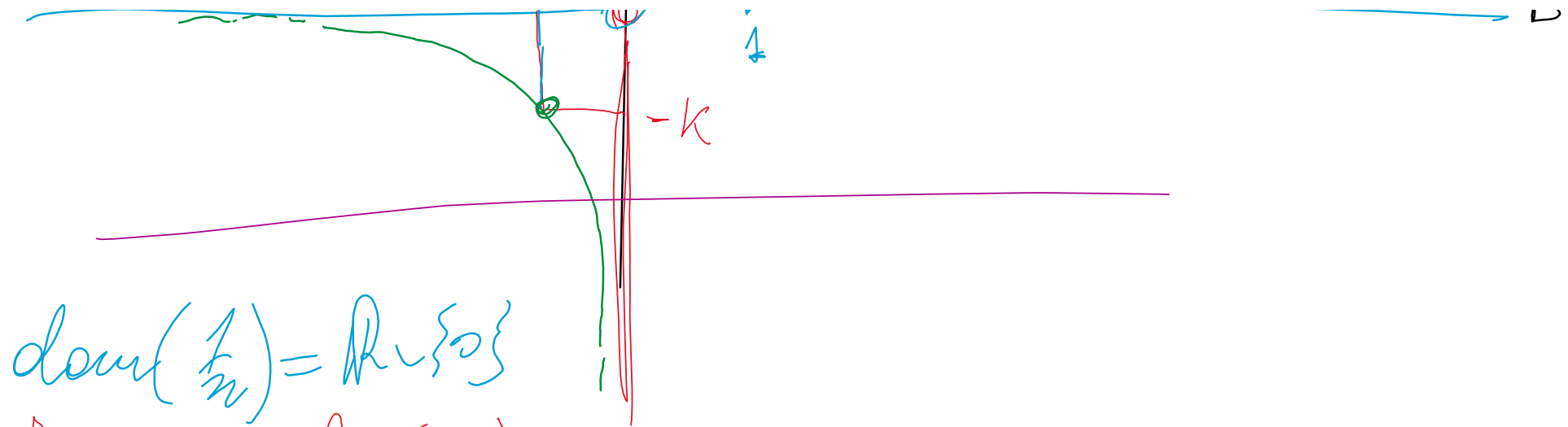






$$f(x) = \frac{1}{x} : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$$





$$\text{dom}(f_n) = \mathbb{R} \setminus \{0\}$$

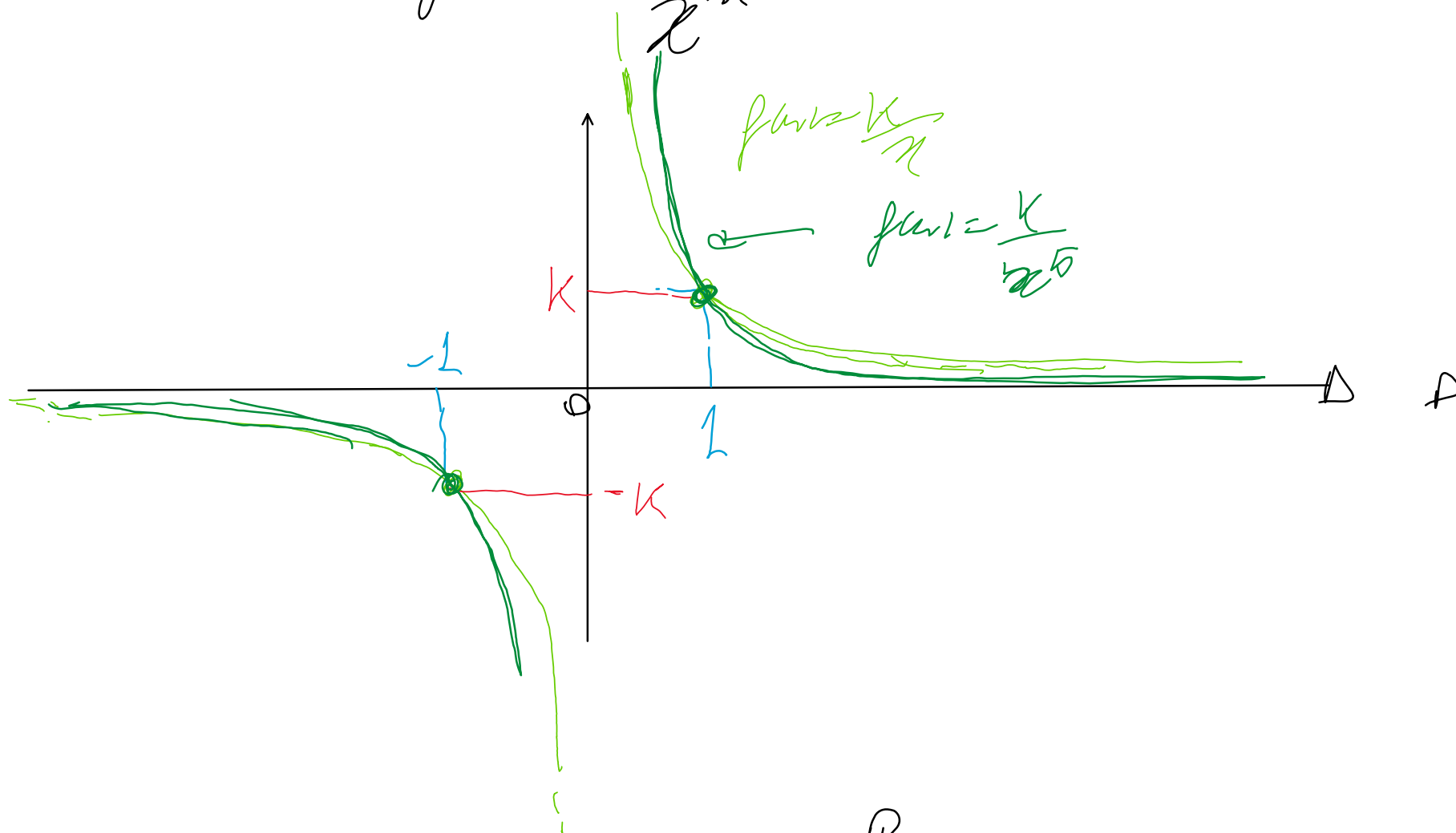
$$\text{Im}(f_n) = \mathbb{R} \setminus \{0\}$$

f non è monotona in $\mathbb{R} \setminus \{0\}$
 ma ha intervalli di STRETTA MONOTONIA

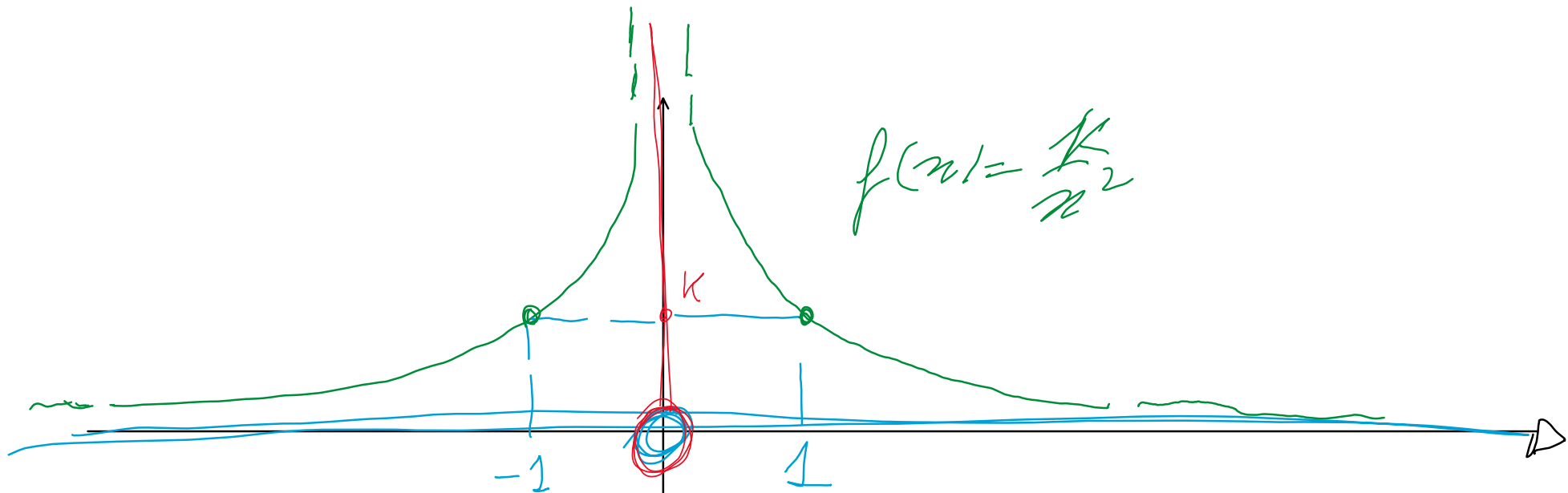
f è INIETTIVA

Ex Trovare la funzione inversa di $f(x) = \frac{1}{x}$

$$f(x) = \frac{K}{x^n} \quad n \text{ DISPARI}$$



$$f(x) = \frac{K}{x^n} : K \in \mathbb{R}^+ \quad n \in \mathbb{N} \quad \text{TARI}$$



$$\text{Dom}\left(\frac{1}{x^2}\right) =]0, +\infty[$$

f non è MONOTONA in $\mathbb{R} \setminus \{0\}$

ma ha intervalli di MONOTONIA STRETTA

$\cap \quad \cup \quad \dots \quad \cap \quad \cup$

f non è INIEITL VA IN $\mathbb{R} \setminus \{0\}$